

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent
Kind Code
Date of Patent
Inventor(s)

12405640
B2
September 02, 2025
Grady; John R. et al.

Adaptable housing for light pipes with different light pipe arrangements

Abstract

A light pipe housing for a drive carrier comprises a base structure having a front face, a back face, a first light pipe cavity, and a second light pipe cavity. The light pipe housing comprises a raised ramp extending between the first light pipe cavity and the second light pipe cavity, the raised ramp having a first edge adjacent to the first light pipe cavity and a second edge adjacent to the second light pipe cavity. The housing further comprises a top cover having a cut out portion. The base structure and top cover are in conjunction configured to carry any one of a first light pipe and a second light pipe for the drive carrier. The light pipe housing configured to carry one light pipe at a time. The first light pipe and the second light pipe have mutually different light pipe arrangements.

Inventors: Grady; John R. (Cypress, TX), Allen; Joseph (Tomball, TX), Bala; Sunil Rao Ganta Papa Rao (Cypress, TX), Sauer; Keith (Spring, TX)

Applicant: HEWLETT PACKARD ENTERPRISE DEVELOPMENT LP (Spring, TX)

Family ID: 91951278

Assignee: Hewlett Packard Enterprise Development LP (Spring, TX)

Appl. No.: 18/163935

Filed: February 03, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240264643 A1	Aug. 08, 2024

Publication Classification

Int. Cl.: H04B10/00 (20130101); G06F1/18 (20060101); H05K7/14 (20060101); F21V8/00 (20060101)

U.S. Cl.:

CPC **G06F1/181** (20130101); **H05K7/14** (20130101); G02B6/0001 (20130101)

Field of Classification Search

CPC: G06F (1/181); H05K (7/14); G02B (6/0001); G02B (6/0095); G02B (6/0073); G02B (6/0081); G02B (6/0083); G02B (6/0086); G02B (6/3825)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5876239	12/1998	Morin	439/490	H01R 13/717
5938324	12/1998	Salmon	362/555	G09F 9/305
6095851	12/1999	Laity	439/910	H05K 5/0273
6099152	12/1999	Naganawa	362/489	B60Q 3/64
6174194	12/2000	Bleicher	439/490	H01R 13/7175
6231224	12/2000	Gamble et al.	N/A	N/A
6352446	12/2001	Ezawa	439/490	H01R 13/6691
6457992	12/2001	Posey	439/131	H01R 13/60
7421184	12/2007	Long	439/490	G02B 6/0008
8861972	12/2013	Hosking	398/138	H04B 10/40
9083123	12/2014	Homan	N/A	H01R 13/7175
9281636	12/2015	Schmitt	N/A	H01R 13/7172
9405080	12/2015	Lambourn	N/A	H04Q 1/136
9608377	12/2016	Phillips	N/A	H01R 13/7172
9612391	12/2016	Wu	N/A	G02B 6/4256
9784927	12/2016	Lambourn	N/A	G02B 6/3895
10064305	12/2017	Zhai et al.	N/A	N/A
11200920	12/2020	Chang et al.	N/A	N/A
11774691	12/2022	Wang	340/815.42	G02B 6/4262
11906801	12/2023	Briant	N/A	G02B 6/4268
12000547	12/2023	Lands	N/A	F21V 21/088
2004/0257760	12/2003	Record et al.	N/A	N/A
2005/0254257	12/2004	Long	362/581	H01R 13/659
2010/0111476	12/2009	Shirk	385/53	H01R 13/7172
2011/0230065	12/2010	Bergner	439/55	G02B 6/426
2014/0023326	12/2013	Anderson	385/77	G02B 6/3895
2014/0023328	12/2013	Lambourn	385/92	G02B 6/42
2019/0296493	12/2018	Liu	N/A	G02B 6/0001
2021/0364687	12/2020	Wang	N/A	G02B 6/4262
2022/0342463	12/2021	Norton et al.	N/A	N/A
2024/0264643	12/2023	Grady	N/A	H05K 7/14

Primary Examiner: Bello; Agustin

Attorney, Agent or Firm: Hewlett Packard Enterprise Patent Department

Background/Summary

BACKGROUND

(1) Electronic devices (such as computing systems) may include one or more media drives such as, solid state drives (“SSDs”), hard disk drives (“HDDs”), CD ROM drives, or DVD drives. One or more receiving locations (e.g., drive bays) can be included in the chassis of an electronic device for receiving these media drives. The media drives (drives) can either be directly mounted within those receiving locations, or may be first received in a drive carrier, which is itself receivable in the receiving location. A drive carrier may be used, for example, in a computing system (e.g., a server, a high-performance-compute system, a data storage appliance, a converged or hyperconverged system, or other computing systems) to facilitate installation and uninstallation (e.g., hot-swapping) of a media drive to the computing system.

(2) Drive carriers are used to configure media drives, such as SSDs, as Field Replaceable Units (FRUs), allowing the drives to be hot swapped into a computing system. Drive carriers are therefore defined by the physical form of the SDD, or drive, which they are configured to carry and are standardized by an industry form factor (e.g., an SDD form factor). An Enterprise & Data Center Standard Form Factor (EDSFF), for example, is designed for use in data centers in servers with storage devices such as SSDs. As a form factor, it defines specifications such as the dimensions and electrical interfaces storage devices should have, to ensure data center operators, server manufacturers, and SSD manufacturers, can make products that work with products from multiple manufacturers. As used herein, an EDSFF drive therefore refers to a drive that has a form factor specified in one of the standards in the EDSFF family of standards, including, but not limited to, a form factor specified in SFF-TA-1006, SFF-TA-1007, or SFF-TA-1008, and an EDSFF drive carrier is a drive carrier configured to carry and support an EDSFF drive. Since different drives, may have different physical dimensions, the drives may also have different card edge placements, different mounting hole locations, and different LED indicator placements. Thus, different drive carriers are used for different drives having different physical dimensions.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present disclosure is best understood from the following detailed description when read with the accompanying figures, wherein:
- (2) FIG. 1A is an exploded view of a light pipe housing with a first light pipe according to an example of the present subject matter;
- (3) FIG. 1B is an exploded view of the light pipe housing of FIG. 1A with a second light pipe;
- (4) FIG. 2A is a front perspective view of the light pipe housing of FIG. 1A with a first light pipe installed therein;
- (5) FIG. 2B is a rear perspective view of the light pipe housing of FIG. 1A with a first light pipe installed therein;
- (6) FIG. 3A is a front perspective view of the light pipe housing of FIG. 1B with a second light pipe installed therein;
- (7) FIG. 3B is a rear perspective view of the light pipe housing of FIG. 1B with a second light pipe installed therein;
- (8) FIG. 4 is an isometric view illustrating a drive carrier frame according to an example of the present subject matter;
- (9) FIG. 5A is a front enlarged partial view of the drive carrier frame of FIG. 4;
- (10) FIG. 5B is a rear enlarged partial view of the drive carrier frame of FIG. 4;

- (11) FIG. 6A is a front enlarged partial view of the drive carrier frame of FIG. 4 fitted with the light pipe housing of FIG. 1A;
- (12) FIG. 6B is a rear enlarged partial view of the drive carrier frame of FIG. 4 fitted with the light pipe housing of FIG. 1A; and
- (13) FIG. 7 is a method of installing a light pipe housing in a drive carrier, according to an example implementation of the present subject matter.
- (14) It is emphasized that in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion or illustration.

DETAILED DESCRIPTION

- (15) Illustrative examples of the subject matter claimed below are disclosed. In the interest of clarity, not all features of an actual implementation are described in this specification. It will be appreciated that in the development of any such actual implementation, numerous implementation-specific decisions may be made to achieve the developers' specific goals, such as compliance with system-related and business-related constraints, which will vary from one implementation to another. Moreover, it will be appreciated that such a development effort, even if complex and time-consuming, would be a routine undertaking for those of ordinary skill in the art having the benefit of this disclosure.
- (16) As used herein, the article “a” is intended to have its ordinary meaning in the patent arts, namely “one or more.” Herein, the term “about” when applied to a value generally means within the tolerance range of the equipment used to produce the value, or in some examples, means plus or minus 10%, or plus or minus 5%, or plus or minus 1%, unless otherwise expressly specified. Further, herein the term “substantially” as used herein means a majority, or almost all, or all, or an amount with a range of about 51% to about 100%, for example. Moreover, examples herein are intended to be illustrative only and are presented for discussion purposes and not by way of limitation.
- (17) As used herein, to “provide” an item means to have possession of and/or control over the item. This may include, for example, forming (or assembling) some or all of the item from its constituent materials and/or, obtaining possession of and/or control over an already-formed item.
- (18) Indicator LEDs are generally used to indicate the state of a drive. A drive may have an activity indicator LED and a status indicator LED. The activity indicator LED may flash whenever the drive is accessed (generally for read/write operations). The status indicator LED may indicate different states of operation of the drive. For example, the status indicator LED may indicate when the drive is not yet discovered by the system, when the drive is online, when the drive is being prepared for removal, or when the drive is subjected to a failure. Different drives, having different physical dimensions, may have different indicator LED placements. In particular, the location of a drive indicator LED for an SSD conforming with E1.S EDSFF specification may be different from the location of a drive indicator LED for an SSD conforming with E3.S EDSFF specification. Due to this difference in location of the indicator LEDs, different light pipe designs are used for different drives conforming with different form factor specifications. To support these different light pipe designs, designs of drive carriers for different drives vary. Designing, tooling, manufacturing, and installing these different drive carriers for different drives involves additional time and cost. Further, stocking of differently designed drive carriers and management of such stocks also adds to additional effort and cost for inventory management.
- (19) The subject matter disclosed herein provides an adaptable light pipe housing for differently designed light pipes, so that the light pipe housing (that is able to hold light pipes of varying designs) may be fitted in a common drive carrier for different drives having varying form factor specifications. In some examples, the present subject matter provides a common drive carrier to be used for EDSFF drives having any one of E1.S or E3.S form factor specifications. Since, the light pipe housing installed within the drive carrier may house light pipes of varying designs, any light

pipe may be fitted within the housing depending on which EDSFF drive, E1.S or E3.S, is to be installed in the drive carrier. By having the common light pipe housing fitted in the carrier, a common drive carrier may be used for multiple drives having varying form factor specifications instead of using uniquely designed drive carriers for each type of drive. By utilizing a common drive carrier fitted with the adaptable light pipe housing, development and production of the drive carriers may be simplified and may be cost effective.

(20) Examples of adaptable light pipe housings are described herein. The light pipe housings are adaptable to accommodate a plurality of differently designed light pipes meant for different EDSFF drives having differing dimensions such as are found in varying specifications for EDSFF form factors. In some examples, the light pipe housing may comprise a base structure and top cover that in conjunction are configured to carry any one of a first light pipe and a second light pipe for a drive carrier, where the first light pipe and the second light pipe have mutually different light pipe arrangements. In an example, the design of the light pipe housing is such that it can hold a light pipe for an EDSFF drive having E1.S specification as well as it can also hold a light pipe for an SSD conformant with E3.S specification, where the light pipe housing is configured to carry one light pipe at a time. Since, the housing is inserted in a common drive carrier, a common drive carrier may be used for two different EDSFF drives conformant with different (E1.S or E3.S) form factor specifications. Thus, by utilizing a common drive carrier fitted with the adaptable light pipe housing, development and production of drive carriers and light pipe housings may be simplified and may be cost effective.

(21) FIG. 1A is an exploded view illustrating a light pipe housing **100** according to an example of the present subject matter. The light pipe housing **100** may hold a light pipe **102** designed to fit with an EDSFF drive having E3.S form factor specifications. The drive may have a first indicator light-emitting diode (LED) on a front face thereof for indicating power and activity of drive. The drive may have a second indicator LED on the front face for indicating operation or error conditions for the drive. In an example, the light pipe housing **100** may be fitted in a drive carrier for the drive.

(22) The light pipe housing **100** has a two-piece construction that includes a base structure **104** and a top cover **106**. The base structure **104** and the top cover **106** may be snap-fitted to each other. The base structure **104** and the top cover **106** are in conjunction configured to carry the light pipe **102**. The light pipe **102** is also referred to as the first light pipe **102**. In some examples, the base structure **104** and the top cover **106** are in conjunction configured to carry a second light pipe **108**, as shown in FIG. 1B. The second light pipe **108** is designed to fit with an EDSFF drive having E1.S form factor specifications. The light pipe housing **100** is configured to carry one light pipe at a time. The first light pipe **102** and the second light pipe **108** have mutually different light pipe arrangements configured to receive light from mutually different LED arrangements associated with different form factors of drives. That is, in some examples, the first light pipe **102** is designed to be utilized with a drive compliant with E3.S form factor specifications which has two LEDs arranged in horizontal arrangement (e.g., at two predetermined locations spaced apart along a horizontal line), whereas, the second light pipe **108** is designed to be utilized with a drive compliant with E1.S form factor specifications which has two LEDs arranged in a vertical arrangement (e.g., at two predetermined locations spaced apart along a vertical line).

(23) As shown in FIG. 1A, the light pipe **102** has a pair of elongate light conductor portions **102a** and **102b**. The light conductor portion **102a** extends between a light receiving end **110** and a light transmitting end **112** and the light conductor portion **102b** extends between a light receiving end **114** and a light transmitting end **116**. When light pipe housing **100** and the light pipe **102** are installed in a drive carrier, the light receiving ends **110** and **114** are disposed near a rear side of a front rail (e.g., front rail **406** described below) so as to be adjacent to (e.g., directly in front of) the LED locations of a drive carried by the drive carrier, wherein the drive has a first LED light arrangement, for example an arrangement comprising horizontally aligned LEDs. Referring to FIG.

1B, the second light pipe **108** has a generally Y-shaped configuration including first and second elongate light conductors **108-a** and **108-b**. The first light conductor **108-a** extends between light receiving end **118** and light transmitting end **120** and the second light conductor **108-b** extends between light receiving end **122** and light transmitting end **124**. When light pipe housing **100** and the second light pipe **108** are installed in a drive carrier, the light receiving ends **118** and **120** are disposed near a rear side of a front rail (e.g., front rail **406**) so as to be adjacent to (e.g., directly in front of) the LED locations of a second type of drive carried by the drive carrier, wherein the second type of drive has a second LED light arrangement different, for example an arrangement comprising vertically aligned LEDs.

(24) In FIGS. **1A** and **1B**, the first and second light pipes **102a** and **102b** that may be supported within housing **100** are configured to be compatible with the indicator LED positioning of EDSFF drives having E3.S form factor specifications and the first and second light pipes **108-a** and **108-b** that may be supported within housing **100** are configured to be compatible with the indicator LED positioning of EDSFF drives having E1.S form factor specifications. In particular, light receiving end **110** of first light pipe **102a** and light receiving end **114** of second light pipe **102b** are oriented to be proximal to the indicator LEDs of E3.S EDSFF drives, and thus the light receiving ends **110** and **114** are horizontally aligned with one another at a spacing and location congruent with the LEDs of the E3.S EDSFF drives while coupled to the drive carrier. On the other hand, while light receiving end **118** of first light pipe **108-a** and second light receiving end **122** of second light pipe **108-b** are oriented to be proximal to the indicator LEDs of E1.S EDSFF drives, and thus the light receiving ends **118** and **120** are horizontally aligned with one another at a spacing and location congruent with the LEDs of the E1.S EDSFF drives while coupled to the drive carrier. Thus, regardless of whether a drive carrier has an E1.S EDSFF drive or an E3.S EDSFF drive installed therein, the adaptable housing **100** according to examples herein may hold either a first light pipe **102** with two elongate light conductor portions **102a** and **102b** or a second light pipe **108** with two elongate light conductor portions **108-a** and **108-b**. The first light pipe **102** and the second light pipe **108** are adapted to redirect light from the indicator LEDs of the EDSFF drives to their respective light transmitting ends, viz., light transmitting ends **112** and **116** for the first light pipe **102** and light transmitting end **120** and **122** for the second light pipe **108**. Further, the spacing and orientation of light transmitting ends **112**; **116** and **120**; **124** on a front face **126** of the base structure **104** is the same regardless of whether an E1.S EDSFF drive or E3.S EDSFF drive is installed.

(25) Referring back to FIG. **1A**, the base structure **104** has a back face **128** opposite to the front face **126**. The front face **126** refers to a surface of the base structure **104** facing the user of the light pipe housing **100** when the light pipe housing **100** is mounted in a drive carrier. The base structure **102** also comprises a first light pipe cavity **130** having a first end **130a** at the front face **126** and a second end **130b** at the back face **128** and a second light pipe cavity **132** spaced apart from the first light pipe cavity **130**. The second light pipe cavity **132** has a first end **132a** at the front face **126** and a second end **132b** at the back face **128**. Each of the first and second light pipe cavities **130** and **132** are channels formed on the base structure **104** and are configured to hold a light pipe therein.

(26) The base structure **104** further comprises a raised ramp **134** extending between the first light pipe cavity **130** and the second light pipe cavity **132**. The raised ramp **134** has a first edge **134a** adjacent to the first light pipe cavity **130** and a second edge **134b** adjacent to the second light pipe cavity **132**. Further, as shown in FIG. **1A**, the top cover **106** has a cut out portion **136**. When the top cover **106** is engaged with the base structure **104**, the cut-out portion **136** is positioned above the second edge **134b** of the raised ramp **134** and the second edge **134b** of the raised ramp **134** is parallel to an edge of the cut out portion **136**. The base structure **104** further comprises a slot **138** running through the front face **126** and the back face **128** of the base structure **104**. The slot **138** is configured to receive a ridged extension formed on a base surface of an adaptor formed in a recess in a drive carrier. In various implementations, the base structure **104**, the top cover **106**, and the

subparts/portions therein, are formed from a plastic material. Examples of the plastic material may include polycarbonate and polycarbonate/acrylonitrile butadiene styrene (PC-ABS).

(27) The base structure **104** further comprises a pair of sidewalls **140** and **142** (also referred to as first sidewall **140** and second sidewall **142**) extending between the front face **126** and the back face **128**. The first and second sidewalls **140** and **142** may extend perpendicularly between the front face **126** and the back face **128**. A first locking clip **144** is formed on the first sidewall **140** and a second locking clip **146** is formed on the second sidewall **142**. The top cover **106** has a first groove **148** to engage with the first locking clip **144** and a second groove **150** to engage with the second locking clip **146** when the top cover **106** is fitted on the base structure **104**. The first and second locking clips **144** and **146** may interlock with the first and second grooves **148** and **150** to ensure that the top cover **106** is snap fitted on the base structure **104**. When the top cover **106** is assembled to the base portion **104**, the locking clips **144** and **146** in the base portion **104** may flex to allow the top cover **106** to fit in position on the base portion **104**. In particular, referring to FIG. 1A, the first locking clip **144** may flex in a direction as indicated by arrow T1 and the second locking clip **146** may flex in a direction as indicated by arrow T2. Since the first and second locking clips **144** and **146** are made of plastic material, once the top cover **106** is fitted in position over the base portion **104**, the first and second locking clips **144** and **146** may snap back in an upright position and thereby fit into the first and second grooves **148** and **150** of the top cover **106** to removably fasten the top cover **106** to the base portion **104**.

(28) The base structure **104** also has a locking receptacle **152**. The locking receptacle **152** is formed on the first sidewall **140** of the base structure **104**. Although not shown in the figures, a locking receptacle similar to the locking receptacle **152** may also be formed on the second sidewall **142**. Thus, a locking receptacle may be formed on each of the pair of sidewalls **140** and **142**. Each of the locking receptacles are configured to receive a respective flange formed on an adaptor in a body of a drive carrier (shown in FIGS. 5A and 5B).

(29) Further, the top cover **106** comprises a first retaining wall portion **154** and a second retaining wall portion **156**. When the light pipe housing **100** is inserted in a recess of a drive carrier frame (FIGS. 5A and 5B), the first and second retaining wall portions **154** and **156** may move inside respective wall portion receptacles formed in the drive carrier frame (shown in FIGS. 5A and 5B). A wall portion receptacle is formed on each side of the recess in the drive carrier frame to receive one of the first and second retaining wall portions **154** and **156**. Once, the retaining wall portions **154** and **156** are inside the receptacles, the light pipe housing **100** may be prevented from sliding out of the recess.

(30) FIGS. 2A and 2B illustrate front and rear perspective views of the light pipe housing **100** of FIG. 1 with the first light pipe **102** installed therein. Referring to FIG. 2A, the top cover **106** is fitted to the base structure **104** and the first light pipe **102** is held therebetween. The light transmitting end **112** of the first light conductor portion **102a** of the first light pipe **102** protrudes out of the first end **130a** of the first light pipe cavity **130**. The light transmitting end **116** of the second light conducting portion **102b** of the first light pipe **102** protrudes out of the first end **132a** of the second light pipe cavity **132**. Referring to FIG. 2B, the light receiving end **110** of the first light conductor portion **102a** of the first light pipe **102** is to protrude out of the second end **130b** of the first light pipe cavity **130**. The light receiving end **114** of the second light conductor portion **102b** of the first light pipe **102** is to protrude out of the second end **130b** of the second light pipe cavity **132**.

(31) FIGS. 3A and 3B illustrate front and rear perspective views of the light pipe housing **100** of FIG. 1 with the second light pipe **108** installed therein. The top cover **106** is fitted to the base structure **104** and the second light pipe **108** is held therebetween. As shown in FIG. 3A, the light transmitting end **120** of the first light conductor portion **108a** of the second light pipe **108** protrudes out of the first end **130a** of the first light pipe cavity **130**. The light transmitting end **124** of the second light conductor portion **108b** of the second light pipe **108** protrudes out of the first end **132a**

of the second light pipe cavity **132**. The light receiving end **118** of the first light conductor portion **108-a** of the second light pipe **108** is to protrude out the cut out portion **136**. At least a portion of the first light conductor portion **108-a** is configured to be held between an edge of the cut out portion **136** and an edge of the raised ramp (not shown in FIGS. 3A and 3B), such that the light receiving end **118** of the first light conductor portion **108-a** of the second light pipe **108** protrudes out of the cut out portion **136**. The light receiving end **122** of the second light conductor portion **108-b** of the second light pipe **108** is to protrude out of the second end **132b** of the second light pipe cavity **132**. The thickness of the base portion **104** is “W” millimeters (mm). In some examples, the thickness ‘W’ may range between 5.5 mm to 7.5 mm.

(32) FIG. 4 is a top isometric view illustrating a drive carrier **400** also referred to as a “drive carrier frame **400**” or “frame **400**.” As shown in FIG. 4, the frame **400** comprises a hybrid, two-piece construction including a first frame section **402** that is made from a metal material and a second frame section **404** that is made from a plastic material. The first and second frame sections **402** and **404** are connected, such that together they form a structure configured to receive and carry a media drive for an electronic device (e.g., a computing device), including, but not limited to an SSD. In one implementation, as described further below, the frame **400** is configured to receive and carry an EDSFF drive. It should be understood, however, that FIG. 4 is not intended to illustrate specific shapes, dimensions, or other structural details accurately or to scale, and that implementations of the frame **400** may have different numbers and arrangements of the illustrated components and may also include other parts that are not illustrated. Furthermore, it should be understood that the disclosed frame **400** may be used to carry various types and configurations of media drives, having different shapes, dimensions, and features, and is not intended to be restricted to a particular drive technology.

(33) The frame **400** includes a front rail **406** comprising a first end portion **405**, a second end portion **407**, and a body **408** extending along a transverse direction T between the first and second end portions **405** and **407**. The frame **400** also includes a first side rail **410** extending from the first end portion **405** of the front rail **406** along a longitudinal direction L perpendicular to the transverse direction T and a second side rail **412** extending from the second end portion **407** of the front rail **406** along the longitudinal direction L perpendicular to the transverse direction T and parallel to the first side rail **410**. In this manner, the front rail **406** in conjunction with the first and second side rails **410** and **412** together form a frame **400** that is configured to receive a substantially cuboid shaped drive (in an open space defined between the first and second side rails **112** and **114**) and to carry the drive for insertion into a substantially cuboid shaped drive bay. The body **408** of the front rail **406**, for example, includes a front face **414** and a back face **416**, and in a state of the drive mounted to the frame **400** the back face **416** is adjacent to and faces the drive. The first and second side rails **112** and **114** are configured to slide into the drive bay, and in an installed state of the frame **400** in the drive bay the front face **414** is accessible to a user.

(34) The front rail **406** comprises a recess **418**. The recess **418** is a through hole formed in the body **408** of the front rail **406**. The recess **418** is configured to receive and carry a light pipe housing, such as the light pipe housing **100** illustrated in FIGS. 1 to 3B. A ridged extension **420** (seen in the enlarged view of FIG. 5A) is formed on the body **408** of the front rail **406**. The base portion **104** of a light pipe housing **100** has the slot **138** (shown in FIGS. 1 to 3B) configured to receive the ridged extension **420**. The light pipe housing **100** (fitted with first light pipe **102** or the second light pipe **108**) may be inserted into the recess **418** along the longitudinal direction L. When the light pipe housing **100** is inserted into the recess **418**, the slot **138** may slide against the ridged extension **420**. It may be noted that if the light pipe housing **100** can be inserted into the recess **418** once the slot **138** is aligned with the ridged extension **420**. Thus, the ridged extension **410** allows for keying the light pipe housing **100** into the recess **418**.

(35) FIG. 5A is a front enlarged partial view of the drive carrier frame **400** of FIG. 4. A portion of the front rail **406** is shown in FIG. 5A. A first adaptor **422** is formed on the body **408** of the front

rail **406**. The first adaptor **422** forms a sidewall of the recess **418** extending opposite to the longitudinal direction L. The first adaptor **422** comprises a first flange **424**. When the light pipe housing **100** is inserted in the recess **418**, the first flange **424** may slide in a locking receptacle (such as locking receptacle **152** shown in FIGS. **1** to **3B**).

(36) FIG. **5B** is a rear enlarged partial view of the drive carrier frame **400** of FIG. **4**. The second adaptor **426** is formed on the body **408** of the front rail **406**. The second adaptor **426** forms a sidewall of the recess **418**, opposite to the first adaptor **422**, and extends opposite to the longitudinal direction L. The second adaptor **426** comprises a second flange **428**. When the light pipe housing **100** (FIGS. **1** to **3B**) is inserted in the recess **418**, the second flange **428** may slide in the locking receptacle **152** (FIGS. **1** to **3B**). Thus, each of the first and second flanges **424** and **428** are received by respective locking receptacles formed in the base portion **104** of the light pipe housing **100** (FIGS. **1** to **3B**). Since, the first and second flanges **424** and **428** are locked in their respective receptacles, the light pipe housing is prevented from shifting from its position within the recess **418**. This provides a tight coupling between the light pipe housing **100** and the drive carrier frame **400** thereby allowing users to hot swap the drive carrier frame **400** (installed with the light pipe housing **100**) from drive bays easily without the risk of the light pipe housing getting displaced from its position. When the light pipe housing **100** is inserted in the recess **418**, the adaptors **422** and **426** in the base portion **104** may flex to allow the top cover **106** to fit in position on the base portion **104**. In particular, referring to FIG. **5A**, the first adaptor **422** may flex in a direction as indicated by arrow X1 and the second adaptor **424** may flex in a direction as indicated by arrow X2. Since the first and second adaptors **422** and **426** are made of plastic material, once the light pipe housing **100** is fitted in the recess **418**, the first and second adaptors **422** and **426** may elastically snap back to their original position. As the adaptors **422** and **426** snap back to their original positions, the first and second flanges **424** and **426** are received by respective locking receptacles formed in the base portion **104** of the light pipe housing **100** thereby locking the light pipe housing **100** in the recess **418**. (Further, the frame **400** comprises a first wall portion receptacle **430** and a second wall portion receptacle **432**. The first and second wall portion receptacles **430** and **432** are formed on a portion of the body **408** of the frame **400** that forms a boundary to the recess **418**. When the light pipe housing **100** is inserted in the recess **418**, the retaining wall portions (**154** and **156** shown in FIG. **1A**) may slide into the wall portion receptacles **430** and **432**. In particular, the first retaining wall portion **154** of the light pipe housing **100** may slide into the first wall portion receptacle **430** and the second retaining wall portion **156** of the light pipe housing **100** may slide into the second wall portion receptacle **432** thereby forming a tight fit between the wall portions and respective receptacles. This prevents the light pipe housing **100** to slide out of the recess **418**).

(37) In various implementations, the first frame section **402** includes the first side rail **410** and the second frame section **404** includes both the front rail **406** and the second side rail **412**. Thus, in such implementations, the first side rail **410** is formed from a metal material and the front rail **406** and second side rail **412** are formed from a plastic material. The first side rail **410** may for example, be formed from a die cast zinc, while the front rail **406** and the second side rail **412** may for example, be integrally formed together from a polycarbonate. Furthermore, in some implementations, the front rail **406** and second side rail **412** are formed from polycarbonate/acrylonitrile butadiene styrene (PC-ABS).

(38) FIG. **6A** is a front enlarged partial view of the drive carrier frame **400** of FIG. **4** fitted with the light pipe housing **100** of FIG. **1A**. FIG. **6B** is a rear enlarged partial view of the drive carrier frame **400** of FIG. **4** fitted with the light pipe housing **100** of FIG. **1A**. In FIG. **6A**, the light pipe housing **100** is shown to be fitted with the first light pipe **102**. In some implementations, the light pipe housing **100** fitted with the second light pipe **108** (FIGS. **3A**, **3B**) may be mounted in the drive carrier frame **400**. The light transmitting ends **112** and **116** of the first light pipe **102** protrude out of the first end **130a** of the first light pipe cavity **130** (shown in FIG. **1-3B**). The light receiving ends

110 and **114** of the first light pipe **102** protrude out of the second end **130b** of the first light pipe cavity **130** (shown in FIG. 1-3B). As the light pipe housing **100** is inserted in the recess **418** (FIG. 5A) of the drive carrier frame **400**, the ridged extension **420** occupies the slot **138**.

(39) FIG. 7 is a method **700** of installing a light pipe housing, such as the light pie housing **100** (FIGS. 1 to 3B) in a drive carrier, such as the drive carrier frame **400** (FIGS. 4 to 5B). The method **700** comprises, at step **702**, providing a light pipe housing comprising a base structure. The base structure comprising a front face, a back face opposite to the front face, a first light pipe cavity having a first end on the front face and a second end on the back face, and a second light pipe cavity spaced apart from the first light pipe cavity. The second light pipe cavity has a first end on the front face and a second end on the back face. The base structure further comprises a raised ramp extending between the first light pipe cavity and the second light pipe cavity. The raised ramp has a first edge adjacent to the first light pipe cavity and a second edge adjacent to the second light pipe cavity. The light pipe housing also comprises a top cover having a cut out portion. At step **704**, the method **700** comprises inserting the light pipe housing into a recess in a front rail of a drive carrier, such as the drive carrier frame **400**. The light pipe housing is inserted in the recess such that the front face of the base structure is exposed at a front of the drive carrier. At step **706**, the method **700** comprises providing any one of a first light pipe, such as the light pipe **102** of FIG. 1A or a second light pipe, such as the light pipe **108** of FIG. 1B for the light pipe housing. The light pipe housing is configured to carry one light pipe at a time, where the first light pipe and the second light pipe have mutually different light pipe arrangements.

(40) In some examples, the method **700** may comprise inserting the drive carrier with the drive into a drive bay of an electronic device, such that the location at the front of the drive carrier is exposed at a front of the electronic device. Further, in some examples, during insertion of the light pipe housing into the recess, a ridged extension, such as the ridged extension **420** (FIGS. 4 to 6B), formed on a body of the drive carrier interacts with a slot, such as the slot **138** (FIGS. 1A to 3B and 6A, 6B) running through the front face and the back face of the base structure. Further, during insertion of the light pipe housing into the recess, flanges, such as the flanges **424** and **428** shown in FIGS. 5A and 5B, each formed on an adaptor, such as the adaptors **422** and **426** shown in FIGS. 5A and 5B, interlocks with a respective locking receptable, such as locking receptable **152** shown in FIGS. 1A and 1B, on a sidewall of the base structure of the light pipe housing.

(41) The foregoing description, for purposes of explanation, used specific nomenclature to provide a thorough understanding of the disclosure. However, it will be apparent to one skilled in the art that the specific details are not required in order to practice the systems and methods described herein. The foregoing descriptions of specific examples are presented for purposes of illustration and description. They are not intended to be exhaustive of or to limit this disclosure to the precise forms described. Many modifications and variations are possible in view of the above teachings. The examples are shown and described in order to best explain the principles of this disclosure and practical applications, to thereby enable others skilled in the art to best utilize this disclosure and various examples with various modifications as are suited to the particular use contemplated. It is intended that the scope of this disclosure be defined by the claims and their equivalents below.

Claims

1. A light pipe housing for a drive carrier, comprising: a base structure comprising: a front face; a back face opposite to the front face; a first light pipe cavity having a first end at the front face and a second end at the back face; a second light pipe cavity spaced apart from the first light pipe cavity, the second light pipe cavity having a first end at the front face and a second end at the back face; and a raised ramp extending between the first light pipe cavity and the second light pipe cavity, the raised ramp having a first edge adjacent to the first light pipe cavity and a second edge adjacent to the second light pipe cavity; and a top cover having a cut out portion, wherein the base structure

- and top cover are in conjunction configured to carry any one of a first light pipe and a second light pipe for the drive carrier, the light pipe housing configured to carry one light pipe at a time, wherein the first light pipe and the second light pipe have mutually different light pipe arrangements.
2. The light pipe housing of claim 1, wherein the cut out portion is positioned above the second edge of the raised ramp, wherein the second edge of the raised ramp is parallel to an edge of the cut out portion when the top cover is engaged with the base structure.
 3. The light pipe housing of claim 2, wherein a light receiving end of a light conductor portion of the second light pipe is configured to protrude out of the cut out portion.
 4. The light pipe housing of claim 3, wherein at least a portion of the light conductor portion of the second light pipe is configured to be held between the second edge of the raised ramp and an edge of the cut out portion, such that the light receiving end of the light conductor portion of the second light pipe protrudes out of the cut out portion.
 5. The light pipe housing of claim 2, wherein a light receiving end of a light conductor portion of the first light pipe is to protrude out of the second end of the first light pipe cavity.
 6. The light pipe housing of claim 2, wherein the light receiving end of the light conductor portion, of the second light pipe cavity, protruding out of the cut out portion is positioned above a light receiving end of another light conductor portion, of the second light pipe, protruding out of the second end of the second light pipe cavity.
 7. The light pipe housing of claim 1, wherein the base structure comprises a slot running through the front face and the back face of the base structure, wherein the slot is configured to receive a ridged extension formed on a drive carrier.
 8. The light pipe housing of claim 1, wherein the base structure further comprises: a pair of sidewalls extending between the front face and the back face, wherein each of the pair of sidewalls has a locking clip to interlock in respective grooves in the top cover.
 9. The light pipe housing of claim 8, wherein the base structure further comprises: a locking receptacle formed on each of the pair of sidewalls, wherein the locking receptacle is configured to receive a respective flange formed on an adaptor in a body of a drive carrier.
 10. The light pipe housing of claim 1, wherein the light pipe housing has a breadth of about 5.5 millimeters (mm) to 7.5 mm.
 11. The drive carrier of claim 1, comprising: an EDSFF drive received in the drive carrier; and one of the first light pipe or the second light pipe, wherein the one of the first light pipe or the second light pipe is received in the light pipe housing.
 12. The light pipe housing of claim 1, wherein the first light pipe is compatible with E3.S EDFF drives.
 13. The light pipe housing of claim 1, wherein the second configuration of the light pipe is compatible with E1.S EDSFF drives.
-